Research Report on User Experience and Peer-to-Peer Exchange Proposition to the Duke Dining System

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I. History and Policy Analysis

There are nearly 50,000 universities worldwide, each one with its own strengths; some excel in Science, Technology, Engineering, and Mathematics (STEM), others in the humanities, and some in the fine arts. Yet, regardless of size, discipline, or reputation, nearly all universities share one commonality: food. At most academic institutions, students relocate to live on campus, creating a fundamental need for a structured system to provide them with meals. This need has given rise to the university dining plan—an essential yet often imperfect solution.

University dining plans are a crucial aspect of student life, designed to provide convenient, cost-effective, and often nutritionally balanced meal options. (Struble, 2013). However, for many students, these plans often fail to align with their actual dining habits, leading to inefficiencies in cost and food access. Some students find themselves with excessive meal swipes or dining dollars that go unused, effectively wasting money, while others struggle to stretch their plans to cover their needs, leading to additional out-of-pocket expenses. This disparity highlights a critical issue in the structure and flexibility of dining plans, affecting both financial well-being and overall satisfaction.

The concept of dining plans in higher education has evolved significantly over time. In earlier days of university, meal plans were rigid, offering a set number of meals at designated dining halls. Over the years, universities introduced more flexible options, such as meal swipes, dining dollars, and tiered plans, to accommodate varying student needs (Struble, 2013). Despite these changes, many students still find themselves stuck in plans that do not fit their eating habits or budgets. This report aims to explore user experience research on student dining plans,

identifying key pain points and proposing solutions to create a more adaptable and equitable system.

II. Competitive Analysis

Duke's rival universities have developed innovative meal plan systems that address a lot of the concerns students face like meal plan flexibility, financial value, and accessibility (see figure 1). While Duke has often been ranked as the best university for food in the United States (by the Daily Meal), other institutions seem to have implemented systems which provide students with greater control over their dining habits.

Georgia Tech allows students to adjust or drop out of their meal plan between semesters without incurring a financial penalty, as long as they do so before the new semester begins. This technique ensures that students who may have initially overestimated their meal requirements are not stuck into a financial commitment that no longer fits their lifestyle. Furthermore, students who discover that they need more meal swipes or dining dollars than anticipated have the option to modify their plans before the next term begins. This helps avoid frequent difficulties like food point wastage and unexpected out of pocket spending by allowing students to reassess their dining patterns and make necessary adjustments.

Emory University offers a financial incentive for students to choose a meal plan rather than paying out of pocket. The university's Dooley Dollars system works similarly to dining dollars at other institutions, but with an added benefit: students who purchase meal plans receive a 5% discount at campus dining locations when they use their Dooley Dollar, and their meals are tax-free, saving an additional 8.9% (depending on state). This discount structure effectively

lowers the per meal cost for students who commit to a meal plan, making it a more appealing choice than paying cash or credit at dining locations.

The University of Delaware's meal swipe exchange program allows students to exchange their dining hall meal swipes for meals at select on campus retail dining chains, including Chick Fil-A and Denny's. This gives students more variety of food without needing them to spend more dining dollars or pay out of pocket at retail places. Furthermore, this allows students who may not always have time for a complete sit down meal in a dining hall to still have access to quality food via quicker fast food options. Since students can apply their meal plan to a broader range of dining options, they are less likely to skip meals due to limited availability or long wait times at dining halls.

Michigan State University features a completely unlimited meal plan where students are able to enter dining halls as much as they want without any concern for using up their meal swipes. Unlike typical meal plans, which limit the amount of meal swipes students can use each semester or week, MSU’s unlimited plan allows for constant access to dining halls on campus. This approach provides an inclusive dining experience. This is especially true for students with schedules that fluctuate rapidly or students who require many small meals throughout the day.

Figure 1

	Duke	Georgia Tech	Emory	University of Delaware	Michigan State
Meal plan adjustability mid semester		✅	✅	✅	
Discounted food using dining dollars			✅	✅	
Meal swipe exchange for retail dining locations				✅	
Unlimited meal swipes package		✅	✅	✅	✅

Each of these universities has carefully created a wide ranging approach that tackles key student issues, such as financial effectiveness, adaptability in meal plan participation, or increasing dining choices past typical dining halls. Through the implementation of multiple adjustable meal plans, several financial incentives, and also complete unlimited dining models, these institutions have created dining systems that better align with the quite varied needs of their student populations.

III. Scope of Research

Overall in terms of scope, our UX research delves into the dining experiences of Duke University undergraduate students and their utilization of the university's meal plan system. The central aim is to identify and address challenges students encounter in managing their food points, with particular attention to the dual problems of food point overabundance and deficit of points. The findings of this research will be instrumental in informing the design and development of a student-operated program or software application designed to facilitate the transaction of discounted food points among students, thereby optimizing resource allocation and reducing waste.

Duke University provides a range of meal plan options for its undergraduate students, particularly those residing on campus. These plans typically operate on a "food points" system, a debit-style arrangement where students purchase a set number of points at the beginning of each semester or academic year. These points can be redeemed at various on-campus dining facilities, including dining halls, cafes, markets, and food trucks. This system offers students the flexibility to choose from diverse culinary offerings across campus. However, the structure of these plans

can lead to challenges in managing point balances effectively. (Duke University Dining Services, n.d., Meal Plans).

How It Works:

Upon selecting a meal plan, students have a specified number of food points allocated to their accounts. Students may enjoy meal swipes, used for all-you-can-eat meals at designated dining halls, such as the Marketplace.

Food points function as a form of currency, with each point equivalent to one US dollar. Food points can be used to purchase meals and snacks at various dining locations on campus.(Duke Student Affairs, 2024). Students can choose from different meal plans based on their year and living situation. First-year students, for example, have specific plans that include both meal swipes and food points.(Duke Student Affairs, 2024).

While the meal plans are designed to cover students' dining needs, the predetermined nature of the point allocation can result in disparities between students' actual consumption and the points available.

Limitations of the Current System:

Several limitations characterize the existing Duke University meal plan system, which imposes an inevitable struggle upon the students.

Fixed Point Allocation:

The fixed number of food points allocated per semester makes it difficult for students to adapt to fluctuating dining needs. This rigidity often leads to either exhausting points early or accumulating a surplus. Students are generally unable to modify their meal plan selection after the initial enrollment period. This inflexibility is a key point of dissatisfaction, as noted in

student feedback forums where many students express the desire for greater adaptability. Some students, on the other hand, report running out of food points before the end of the semester, especially if they frequently dine at more expensive locations (Reddit, 2023).

Refund Policy:

Duke University does not provide refunds for unused food points or meal swipes at the end of the semester or academic term (Reddit, 2023).

Point Expiration:

Unused food points typically do not roll over to subsequent semesters. This "use it or lose it" policy has been a frequent topic of discussion in student publications, with concerns raised about the financial loss incurred by students. (The Chronicle (Duke Student Newspaper), 2023).

Potential for Waste:

The combination of surplus points and point expiration incentivizes students to overspend unnecessarily towards the end of the semester, contributing to food waste.

Absence of Peer-to-Peer Exchange:

Currently, there is no established or efficient mechanism for students with excess points to transfer or sell them to students facing a deficit.

Target Personas:

This research focuses on Duke University undergraduate students (both domestic and international) who participate in the university's meal plan program. This diverse population exhibits a range of dining habits, dietary preferences, and financial situations, all of which shape their interactions with the meal plan system. The study considers three primary user personas:

- The “Over-estimator”: Students who purchase a larger meal plan than they need, resulting in a surplus of food points at the end of the semester.
- The “Under-estimator”: Students who underestimate their dining needs, leading to the early exhaustion of their food points.
- The “Balanced User”: Students who accurately estimate their dining needs and effectively utilize their food points.

IV. UX Research Methods

This research employs the following UX methodologies to comprehensively examine the user experience:

- User Interviews: In-depth discussions with students to gather qualitative data about their experiences, opinions, and challenges concerning the meal plan.
- "I Like, I Wish, What If" Exercise: A structured activity to elicit students' positive experiences ("I Like"), areas for improvement ("I Wish"), and suggestions for innovative solutions ("What If")
- The Love/Breakup Letter Exercise: Using each of our target personas, love/breakup letters were drafted to fully flesh out the problem statement. Through what is seemingly a silly exercise, valuable perspectives about the problem are gained.
- Empathy Maps: Consideration of users by diving into how they would perceive the topic we are trying to tackle through things they may say, feel, think, or Do

IV a. The User Interview

User interviews serve as a crucial qualitative research method in UX design, providing rich, detailed insights into user experiences, behaviors, and motivations. Through direct, one-on-one conversations, we gain valuable information that complements quantitative data. In this study, user interviews are used to explore Duke University undergraduate students' perspectives on the meal plan system and their interest in a potential food point exchange program.

The UX user interviews aimed to understand students' lived experiences with the Duke meal plan, focusing on their current meal plan and its adequacy, how often they monitor their food point balance, their willingness to buy or sell food points at a discount, and their preferred platform, pricing, and security measures for a food point exchange system.

The interviews were semi-structured, allowing for flexibility and in-depth exploration of interesting responses. They revealed a strong desire for a more flexible system. The willingness to participate in a discounted exchange program is clear. Key takeaways include the need for app security, varied feelings about daily limits and filtering, and a general acceptance of university oversight. Concerns about pricing and taxes highlight the financial considerations of students, since purchasement from the school requires paying more taxes emphasizing the need for a fair and transparent system.The interviews revealed a strong desire for a more flexible system, and a willingness to participate in a discounted exchange program. Key findings emphasized the need for app security and a general acceptance of university oversight, while opinions on daily limits and filtering varied. Concerns about pricing and taxes highlighted the financial considerations of

students, especially the increased tax burden when purchasing from the school, emphasizing the need for a fair and transparent system.

As a result of these in person interviews, the problem at hand can be broken into two sections: what's working and what is not.

What's Working?

1. **Convenience of the Meal Plan**

   Across all letters, students appreciated the ease of accessing food without having to cook or grocery shop, especially in comparison to students at other universities. The variety of food available on campus, particularly at WU, was seen as a strong positive.

2. **Flexibility in Meal Plan Choices After Freshman Year**

   Sophomores and upperclassmen like having the ability to choose how many food points they want instead of being locked into a fixed plan like freshmen.

   Students enjoy being able to use food points at multiple locations rather than just one dining hall.

3. **Charitable Option (FeedEveryDevil)**

   The option to donate excess food points was viewed positively, particularly by students with too many points.

What's Not Working?

1. **Running Out of Food Points Too Early**

   Freshmen struggle to make their required meal plan last the entire semester,

especially because WU is a more appealing option than Marketplace. Sophomores who underestimate their needs end up in a tough situation, having to pay out of pocket or ration their meals in the last few weeks.

2. **Wasting Food Points at the End of the Year**

   Some students overestimate how much they need and are left with large balances that don't roll over after spring semester. The lack of a way to recover or repurpose these extra points leads to waste. While donating is an option, some students regret not being able to get any value back for their unused points.

3. **Lack of a Middle Ground Between Too Many and Too Few Food Points**

   There is no way for students to adjust their food point balances mid semester, those who run out early are stuck, and those with too many have to find creative ways to spend them before they expire. The current system relies on students predicting their food habits for an entire semester or year, which is difficult.

IV b. "I Like, I Wish, What If"

This exercise offers a unique and interactive approach to user interviews. Using this UX research method, participants are gathered in a room and presented with three prompts: **"I Like…"**, **"I Wish…"**, and **"What If…"**. They are encouraged to complete each statement by jotting down as many responses as they can on sticky notes, which are then placed under the corresponding category. The most common responses from each prompt were synthesized, consolidated, and summarized (see figure 2).

Figure 2

I Like	I Wish	What If
• The convenience of mobile order • Being able to use food points at the Lobby Shop, vending machines, and Food Trucks • Having the option to pick different plan options • Being able to track my points	• The points would roll over by year • I could simply pay out of pocket for what I need instead of a fixed amount • There was more transparency in taxing • I could use/transfer my food points for other on campus purchases • I could customize how many food points to load onto my account • I didn't have to waste my points • I didn't always run out	• The money I spend on food points is liquid/transferrable? • I could make money off of the money I lose? • I didn't lose value from my money? • I didn't have to worry about having too many points? • I didn't have to worry about running out?

IV c. The Love and Breakup Letter

The "Love/Breakup Letters" method is a creative user experience research technique that invites participants to personify their relationship with a product or service by writing a letter expressing affection or ending the relationship. This approach allows users to articulate their emotional responses, uncovering deep insights into their experiences and perceptions. In our study, we utilized this method by drafting such letters for each of our target personas. Although this exercise may seem unconventional, it effectively reveals valuable perspectives on the problem space, highlighting both positive aspects and pain points that might not surface through traditional research methods (See figure 3).

Figure 3:

Duke Freshman

Class of 2028

freshman@duke.edu
+123-456-7890
Durham, NC

Letter 1: Freshman (International Student)

Dear Duke Meal Plan,

I mean, it's not terrible. Actually, it's kind of nice. Coming from another country, I thought I'd have to struggle with food, but nope, I just swipe my card and get meals. No cooking, no grocery shopping, no stress. My friends studying in other countries have to cook all the time, and I feel lucky I don't have to.

But also, how do I run out of food points so quickly? I told myself I'd be smart, only use food points for lunch and eat Marketplace for dinner, but Marketplace food just doesn't hit the same. So obviously, I ended up eating dinner at WU way more than I was supposed to. And now? It's not spring break yet, and I'm almost broke. I either eat Marketplace for every meal or will have to start paying out of pocket, which sucks because I already paid for this meal plan.

I just wish it lasted longer. That's all. I like not having to worry about food, but now I do have to worry about food, so yeah. Not sure what I'm supposed to do differently next time.

Duke Freshman

Duke Sophomore

Class of 2027

sophomore@duke.edu
+123-456-7890
Durham, NC

Dear Duke Dining,

I don't know how this happened, but I have way too many food points and zero chance of using them all before the semester ends. I was trying to be responsible, making sure I had enough to last, but now I'm out here buying overpriced coffee and grabbing snacks for friends just to get rid of them.

I'll probably donate a bunch to Feed Every Devil since I don't want them to go to waste. That's nice and all, but honestly, it kind of sucks that I spent all this money and won't get any of it back. I wish I had a better idea of how much I actually needed before I loaded up my plan. Now I know for next year, but still.

The food itself is very good and I have no complaints, and I like not having to cook. I just wish I hadn't overestimated how much I'd eat. Definitely going to be more careful next time.

Duke Sophomore

Duke Off-Campus Senior

Class of 2025

senior@duke.edu
+123-456-7890
Durham, NC

Dear Meal Plan,

We had a good run, but I think we're done.

I get why I needed a meal plan when I lived on campus, super convenient, no thinking required. But now that I'm off campus? Not worth it. I cook most of my meals, grab lunch at WU when I feel like it, and just pay with my card. It's honestly easier that way.

I thought about getting a meal plan this year, but why would I? A food point is literally just a dollar, so I'd rather just pay as I go. If I bought a plan, I'd have to worry about running out or having too many, and I don't have time for that.

So yeah, meal plan, it's been real, but I think I'm good. Appreciate the memories though.

Duke Sophomore

Duke Sophomore #2

Class of 2027

sophomore@duke.edu
+123-456-7890
Durham, NC

Dear Duke Dining,

Yeah, so I'm basically out of food points, and I don't know what to do. I thought I had plenty at the start of the semester, but somehow they just disappeared? Like, I wasn't even being that reckless, I ate most of my meals at WU like a normal person, got coffee, maybe a few late night snacks. But now I check my balance and it's definitely not enough for the semester.

So now I either start paying out of pocket, which I really don't want to do, or I just eat less? I don't even know. I already cut out the dumb little impulse buys, but even normal meals are draining the last of what I have. I wish I could just stretch it a little further.

Anyway, guess I'll be budgeting better next semester. But for now, I'm about to start living off granola bars.

Duke Sophomore who Miscalculated

IV d. Empathy Maps

Empathy maps are another valuable UX research tool by helping us to gain a deeper understanding of our users by stepping into their shoes. This method involves analyzing how users might perceive the issue at hand by considering four key aspects of their experience: what they say, feel, think, and do. By mapping out these elements, we can uncover emotional motivations, pain points, behavioral patterns that influence how users interact with the system, and better discover the job to be done. For example, a student running out of food points might say they feel frustrated with the lack of flexibility, think that meal plans are overpriced, feel stressed about affording extra meals, and do things like skipping meals or borrowing points from friends. On the other hand, a student with excess food points might say they don't want their parents' money to go to waste, think the university should offer rollover options, feel annoyed at the rigidity of the system, and do things like buying excessive snacks just to use up their balance. By visualizing these perspectives, empathy maps allow us to better design a solution that meets users' real needs and emotional concerns. (See Figure 4 and 5)

Figure 4

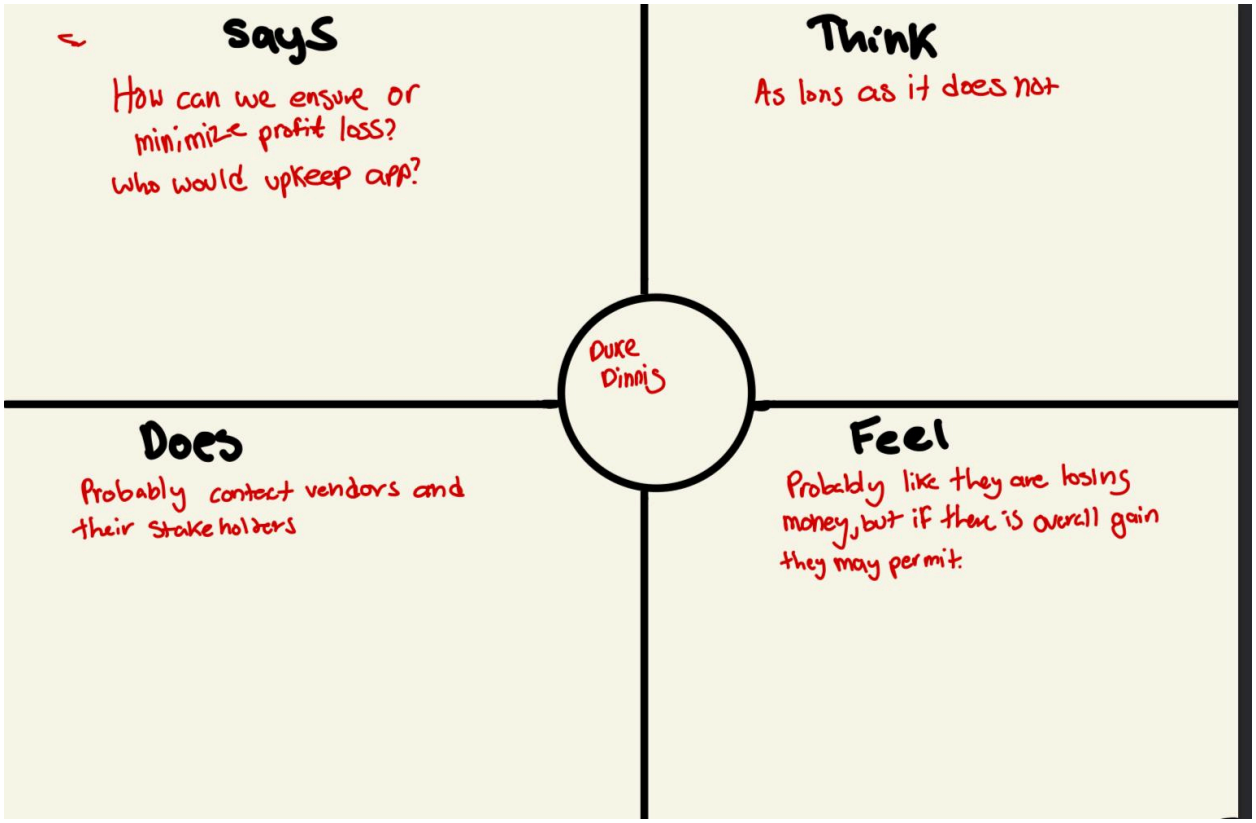
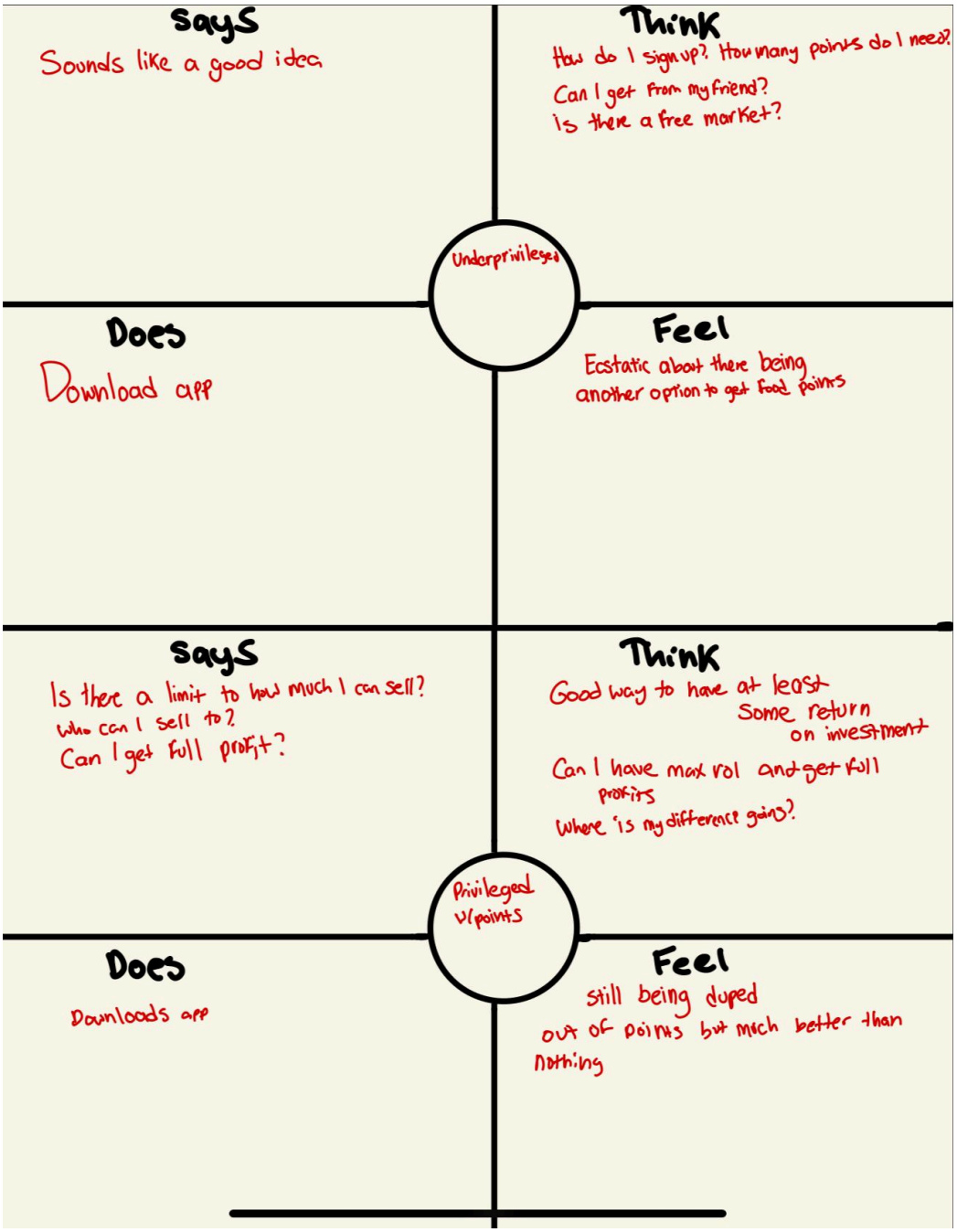

Says
How can we ensure or minimize profit loss?
Who would upkeep app?
Think
As lons as it does not
Duke Dinnis
Does
Probably contect vendors and their stakeholders
Feel
Probably like they are losing money, but if there is overall gain they may permit.

Figure 5


Says
Sounds like a good idea
Think
How do I sign up? How many points do I need?
Can I get from my friend?
Is there a free market?
Underprivileged
Does
Download app
Feel
Ecstatic about there being another option to get food points
Says
Is there a limit to how much I can sell?
Who can I sell to?
Can I get full profit?
Think
Good way to have at least some return on investment
Can I have max vol and get full profits
Where is my difference gain?
Privileged w/points
Does
Downloads app
Feel
still being duped out of points but much better than nothing

V. User Personas

User-Persona 1: The "Under-estimator" (See Figure 6)

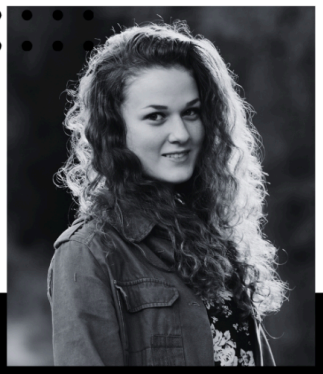
Sarah is a sophomore at Duke University, majoring in Economics with a minor in Public Policy. Coming from a middle-income family, she carefully manages her finances and sticks to a strict budget. She relies heavily on her university meal plan to cover her dining expenses, as she does not have the financial flexibility to waste food points or spend extra money on meals. At the beginning of each semester, she purchases a dining plan that seems reasonable, but by mid-semester, she often finds herself running low on food points. Meanwhile, some of her friends end up with an excess balance, unsure how to use their remaining points before they expire.

By mid-November, Sarah realizes she only has a few food points left, and she cannot afford to keep paying full price for meals out-of-pocket. She starts cutting back on dining hall visits, but this is stressful and inconvenient. Then, she hears about the peer-to-peer food point exchange system, where students with excess points can sell them at a discounted rate. She logs into the platform, finds a seller, and purchases additional points at a better deal than the university's refill rates. This system not only helps her stay within her budget but also ensures another student can recoup some value from their unused points.

By the end of the semester, Sarah is relieved that she doesn't have to go hungry or spend money she doesn't have. She tells her friends about the platform and plans to use it again next

semester—whether to buy points when needed or sell any extra she might have. For her, this system isn't just convenient—it's a financial necessity.

Figure 6



SARAH

AGE: 20

Sophomore
Middle/Low class - cannot afford to be wasting money

GOALS

- Find a way to buy food points at a discounted rate to stretch her budget further.
- Ensure that students with extra food points don't waste their remaining balance and can recover some value.
- Use a safe and university-backed platform instead of relying on informal or unreliable methods.
- Have a more flexible and student-friendly meal plan system that allows for better resource distribution.

FRUSTRATIONS

- Running out of food points before the semester ends and being unable to afford additional meals at full price.
- Inflexibility of current dining plans, which don't allow students to easily adjust their balance based on real usage.
- Financial stress from having to pay for food out-of-pocket when her meal plan runs low.
- Wasted meal plan funds when other students have excess food points that could be redistributed.
- Lack of an official, structured system to facilitate secure and fair food point exchanges.

A Peer-to-Peer Exchange sysetm can solve her problem: allow her to buy the smallest food plan, allow her to buy additional food points at a discounted cost that she can afford

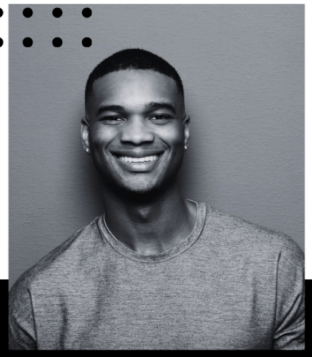
Personal Requirement: The platform is safe and promoted by the University, for reassurance.

User Persona 2: The “Over-estimator” (See Figure 7)

David is a junior at Duke University, majoring in Computer Science with a certificate in Innovation and Entrepreneurship. He comes from a financially stable background, and his parents prefer to purchase the largest meal plan available to ensure he never runs out of food points. However, David’s eating habits don’t align with his plan—he often skips breakfast, eats off-campus with friends, and prefers cooking simple meals in his dorm. As a result, by the middle of the semester, he notices that his food point balance remains far higher than he could possibly use before the term ends.

As David nears the end of the semester, he realizes he still has hundreds of unused food points. He doesn’t want his parents’ money to go to waste, but he also doesn’t need to buy stacks of bottled drinks or snacks just to use up the balance. He wishes there was a way to sell some of his excess food points to students who need them. That’s when he hears about the peer-to-peer food point exchange system. He logs in, lists a portion of his balance at a discounted rate, and within a day, a student in need purchases them. David gets back some of the money he would have otherwise lost, and the buyer secures food points at a more affordable price. By the end of the semester, David feels relieved—he didn’t waste his food points, he helped another student, and he found an easy, fair way to manage his dining plan more effectively. Next semester, he plans to use the system proactively so he can offload extra points earlier and avoid the last-minute rush to spend them.

Figure 7



DAVID

AGE: 21

Junior
Higher Class - can afford higher plan but is still money conscious

GOALS

- Recoup some of the money spent on unused food points instead of wasting them.
- Help other students by offering a discounted way to access food points.
- Use a convenient, university-supported system that allows for safe and simple transactions.
- Gain flexibility in managing his meal plan rather than being locked into an overly large, non-adjustable plan.

FRUSTRATIONS

- Always having excess food points, leading to wasted money at the end of each semester.
- Feeling pressured to overspend on campus dining or buy unnecessary items to use up points before they expire.
- No efficient way to transfer food points to students who actually need them.
- Frustration with the rigidity of dining plans, which don't adjust based on actual meal habits.

A Peer-to-Peer Exchange sysetm can solve his problem: allow him to recoup his points by selling them at a discounted rate to his peers, while also having more flexibility

Personal Requirement: The platform still allows him to keep his eating habits

User Persona 3: The "Balanced User" (See Figure 8)

Ren is a senior at Duke University, majoring in Public Health with a minor in Statistics. Over the years, she has mastered the art of budgeting her food points, carefully planning her meals so she uses just the right amount by the end of the semester. She enjoys eating at campus dining locations but also balances her spending with occasional off-campus meals. While she usually finishes the semester with little to no leftover food points, there have been times when unexpected changes in her routine—like attending more social dinners or traveling home for a few weekends—have thrown off her carefully planned budget.

As Ren approaches the last few weeks of the semester, she realizes she has only a handful of food points left. Normally, she would try to stretch them by skipping coffee runs or choosing cheaper meal options, but with finals coming up, she knows she'll need convenient and nutritious meals. Instead of paying full price to reload her account or pay out of pocket, she logs into the peer-to-peer food point exchange system, where students with extra points are selling them at a discounted rate. She quickly finds a seller offering points at a lower price, allowing her to stay within budget while still getting the meals she needs.

At the same time, Ren appreciates the system from the other side—if she ever ends up with extra food points in future semesters, she now has a reliable way to sell them and recoup some of her costs instead of wasting them. Even though she usually spends her food points just right, having the flexibility to buy or sell when needed provides her with peace of mind, ensuring she never has to stress about small miscalculations in her meal budget.

Figure 8

REN

AGE: 22

Senior
Middle/Low class - she is an avid budgeter

GOALS

- Have access to a backup plan in case she slightly miscalculates her food point usage.
- Save money when needed by purchasing food points at a discounted rate instead of paying full price.
- Avoid stress about running low near the end of the semester while still maintaining her balanced spending habits.
- Ensure that food points remain a flexible resource rather than a rigid budget constraint.

FRUSTRATIONS

- No safety net if she runs low on food points due to unexpected circumstances.
- Missed opportunities to buy food points at a discounted rate when she occasionally needs more.
- Lack of flexibility in the dining plan, forcing her to make precise calculations each semester.
- Difficulty using up exactly the right amount of food points, leading to small inefficiencies.

A Peer-to-Peer Exchange sysetm can solve her problem: their is more flexibility and a backup plan for the end of the semester so she can ensure that she is never on a rigid budget

Personal Requirement: The platform is secure

VI. Key Insights

User research revealed a strong demand for more flexibility in the system to manage food points, specifically reduction in the amount of food points that go to waste each year. 88.3% of students surveyed and interviewed reported having either an excess or a shortage of points each semester. Many expressed frustration with the rigidity of the current structure, which often results in financial inefficiencies: 50% of respondents specifically stated that they feel their money goes to waste under the existing system. To address this, students overwhelmingly supported a peer-to-peer exchange platform, with 100% of participants indicating they would be willing to buy or sell food points at a discount. With all of these considerations, we can extract actionable key insights.

Financial Considerations

Finance remains a major concern across all interviews and research methods when anything related to the Duke dining plan is discussed—particularly regarding pricing and taxation. Since purchasing food points directly from the university incurs higher tax costs, many students stressed the need for a fair and transparent pricing structure. Our research has found that, due to the taxing on the food plan in addition to tax on individual purchases, students would rather pay out of pocket than upgrade their food plan.

Frustration About Waste

The research furthermore suggests a significant student frustration when it comes to waste. As points do not roll over by year, students are left with no choice but to either over spend

and waste excess points, or choose to spend less than needed while scrambling to find ways to still purchase food.

Security

Security emerged as a top priority among users, who emphasized the need for a safe and reliable platform to facilitate transactions. The university already touts a robust mobile ordering system. But if this system were to be expanded to meet user needs in financial waste, security concerns arise. Concerns about fraud or exploitation suggest that features such as secure payment processing, identity verification, and transaction tracking should be integrated into the app to build trust and encourage participation. A well-structured security framework will be essential to ensuring the platform's long-term success.

University Oversight

University oversight was another key consideration. While students generally accepted that some level of regulation would be necessary, they expressed the need for transparency in how the university would be involved. Some participants worried that excessive oversight might introduce restrictions or fees that would undermine the platform's benefits, while others believed university involvement could help legitimize the system. Striking a balance between autonomy and administrative cooperation will be crucial in ensuring both student satisfaction and institutional approval.

Opinions on daily transaction limits and filtering options were mixed. Some students supported transaction limits as a way to prevent stockpiling or exploitation, while others preferred an unrestricted system for maximum flexibility. Similarly, filtering options—such as sorting offers by price, location, or seller reputation—received varied reactions. Some users saw these features as helpful tools for informed purchasing, while others felt they might add unnecessary complexity. Providing customizable options may allow students to tailor the experience to their preferences.

VII. Discussion

The research findings confirm a strong student interest in reforming the current dining plan system, which underscores the need for a more adaptable and equitable approach. A significant number of students experience inefficiencies in their meal plans, either overpaying for unused swipes or struggling to stretch their dining dollars throughout the semester. Either way, value is being lost, and greater flexibility may allow for students to not lose such value. Indeed, a promising approach is the implementation of a food point exchange system, allowing students to trade excess meal swipes or dining dollars in a structured, university-supported marketplace. However, for such a system to be successful, it must be designed with careful consideration of security, flexibility, and financial fairness. For the system to be most successful, it should allow students to buy and sell food points among each other at a discounted rate, creating a mutually beneficial exchange. The discounted price would serve as an incentive for students who run out of food points, offering them a more affordable alternative to purchasing additional meals outright. Instead of having to pay full price at retail locations or resort to expensive off-campus

dining options, students could replenish their food points at a lower cost, making the system more accessible and budget-friendly. Simultaneously, for those who have excess food points, this system ensures that their unused balance does not go to waste. Rather than losing value at the end of the semester or feeling compelled to overspend just to use up their remaining balance, students could recoup a portion of their investment by selling their surplus points to those in need. This exchange fosters a more efficient and equitable distribution of resources, reducing financial strain while maximizing the utility of dining plans for all students. Research on similar peer-to-peer exchange platforms suggests that clear policies and user-friendly design can mitigate risks while enhancing usability (Carroll, Belotti, Lampinen, Airi, et al.). If implemented thoughtfully, this platform could significantly improve how students manage their food budgets, creating a more efficient, fair, and student-centric dining system. By striking the right balance between accessibility and regulation, universities have the opportunity to modernize their dining services, better aligning them with students' diverse needs and spending habits.

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